

JADAVPUR UNIVERSITY
B.E. ELECTRICAL ENGINEERING EXAMINATION - 2024
ELECTRICAL MACHINES - II (Part I)

Detailed Analytical Solutions provided by SR Sir

Question 1(a): Why an induction machine is called a generalized transformer?

SR Sir's Thought Process

This is a recurring fundamental question. The core similarity is Faraday's Law of Induction. The stator acts as the primary, the rotor as a short-circuited secondary. The presence of the air gap and rotation are the only differences. I need to demonstrate that a static transformer is just a special case of the induction motor where slip $s = 1$.

Answer:

An induction machine is called a *generalized transformer* because its fundamental principle of operation, electromagnetic energy transfer, and equivalent circuit structure are identical to that of a static transformer.

In both devices, power is transferred from the primary winding (stator) to the secondary winding (rotor) entirely through mutual magnetic induction. There is no direct electrical connection between the two circuits. The key features that establish this relationship are:

1. **Flux Excitation:** Both machines require a magnetizing current to set up the mutual working flux.
2. **Frequency Relationship:** In a transformer, primary and secondary frequencies are equal ($f_1 = f_2$). In an induction motor, the rotor frequency depends on the relative speed (slip, s): $f_2 = sf_1$.

If we physically lock the rotor of an induction motor so it cannot spin ($N_r = 0$), the slip becomes $s = \frac{N_s - 0}{N_s} = 1$. Under this standstill condition, the rotor frequency becomes identical to the stator frequency ($f_2 = f_1$), and the motor behaves exactly like a standard short-circuited transformer with an air gap. Because the static transformer is merely a restricted special case (permanently operating at $s = 1$) of the induction machine, the induction motor represents the broader, more "generalized" case of transformer action.

Question 1(b): What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor?

SR Sir's Thought Process

Another classic conceptual test. The primary is now the rotating body, and the secondary (stator) is fixed to the ground. By Lenz's Law, the induced current in the secondary will try to oppose the relative motion. Since the stator can't move, the reaction force pushes the rotor in the opposite direction.

Answer:

If the primary winding is placed on the rotor and supplied with poly-phase currents, while the stator is short-circuited acting as the secondary, the rotor will rotate in the **opposite direction** to the direction of its own rotating magnetic field (RMF).

Reasoning: When the 3-phase supply energizes the rotor, it generates an RMF that revolves at synchronous speed (N_s) relative to the rotor structure itself. This moving magnetic flux cuts across the stationary, short-circuited stator conductors, inducing an EMF and subsequently a current in them.

According to Lenz's Law, the electromagnetic torque developed will act in a direction that opposes the cause producing it—which is the relative motion between the rotating flux and the stationary stator. The magnetic interaction attempts to drag the stator along with the RMF to reduce this relative speed. However, because the stator is physically bolted to the frame and cannot move, the reaction torque acts heavily on the movable component: the rotor. Consequently, the rotor is forced to physically rotate backwards (in the direction **opposite** to the RMF) to try and reduce the absolute speed of the magnetic field cutting the stator.

Question 1(c): What is the relative speed between the magnetic field produced by stator winding and the magnetic field produced by the rotor winding? Give reasons for your answer.

SR Sir's Thought Process

This is identical in spirit to the 2023 1(a) question. I must calculate the absolute spatial speed of both fields independently and show that they are identical (N_s). If their absolute speeds are identical and in the same direction, their relative speed is zero.

Answer:

The relative speed between the magnetic field produced by the stator winding and the magnetic field produced by the rotor winding is **Zero**.

Reasons: To find the relative speed, we must determine the absolute speed of both magnetic fields in space (i.e., with respect to the stationary stator frame).

1. **Stator Magnetic Field:** When a balanced 3-phase supply of frequency f_1 is applied to the stator, it generates a Rotating Magnetic Field (RMF) that revolves at synchronous speed.

$$\text{Speed of Stator Field in space} = N_s = \frac{120f_1}{P} \quad (1)$$

2. **Rotor Magnetic Field:** The rotor physically rotates at a mechanical speed of N_r . The relative speed of the stator field with respect to the rotor conductors is the slip speed ($N_s - N_r = sN_s$). This induces a current in the rotor at slip frequency ($f_2 = sf_1$). These slip-frequency currents produce their own RMF. The speed of this rotor RMF *with respect to the rotor body itself* is:

$$N_{\text{rotor_relative}} = \frac{120f_2}{P} = \frac{120(sf_1)}{P} = sN_s \quad (2)$$

To find the absolute speed of the rotor field in space, we must add the mechanical speed of the rotor body:

$$\text{Speed of Rotor Field in space} = N_{\text{rotor_relative}} + N_r = sN_s + (1 - s)N_s = N_s \quad (3)$$

Since both the stator field and the rotor field are rotating at the exact same synchronous speed (N_s) in the exact same direction, they are magnetically locked together. Therefore, the relative speed between them is exactly zero.

Question 1(d): Find an expression for pitch factor for a short pitched winding.

SR Sir's Thought Process

Note the typo in the question paper ("peach factor" instead of pitch factor). I will address the derivation for Pitch Factor (K_p). A full-pitch coil has coil sides exactly one pole pitch (180° electrical) apart. EMFs add directly arithmetically. A short-pitched coil is shorter by an angle α . The EMFs are out of phase by α . I'll use a simple phasor addition diagram to derive $K_p = \cos(\alpha/2)$.

Answer:

*Instructor's Note: The term "peach factor" in the question paper is a typographical error for **Pitch Factor** (K_p), also known as the coil span factor.*

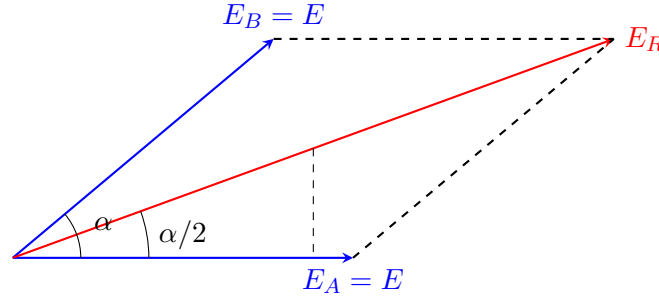
Derivation of Pitch Factor (K_p): Pitch factor is defined as the ratio of the phasor sum of induced EMFs per coil to the arithmetic sum of induced EMFs per coil.

$$K_p = \frac{\text{Phasor sum of coil side EMFs}}{\text{Arithmetic sum of coil side EMFs}} \quad (4)$$

Let us consider a single coil consisting of two coil sides, A and B. If the coil is **full-pitched**, the distance between the coil sides is exactly one pole pitch (180° electrical). The EMFs induced in both sides (E_A and E_B) are perfectly in phase with respect to the coil loop. Thus, if $E_A = E_B = E$, the arithmetic sum is:

$$E_{\text{arithmetic}} = E + E = 2E \quad (5)$$

If the coil is **short-pitched** by an electrical angle α , the coil span is $(180^\circ - \alpha)$. Consequently, the EMF induced in coil side B will lag (or lead) the EMF in coil side A by the angle α . To find the resultant EMF of the coil (E_R), we must add E_A and E_B vectorially.



From the phasor geometry, the resultant E_R bisects the angle α because the magnitudes E_A and E_B are equal. Drawing a perpendicular from the tip of E_A to the resultant bisects E_R .

$$\frac{1}{2}E_R = E \cos\left(\frac{\alpha}{2}\right) \implies E_R = 2E \cos\left(\frac{\alpha}{2}\right) \quad (6)$$

Substitute these back into the definition of Pitch Factor:

$$K_p = \frac{E_{\text{phasor}}}{E_{\text{arithmetic}}} = \frac{2E \cos(\alpha/2)}{2E} \quad (7)$$

$$\boxed{K_p = \cos\left(\frac{\alpha}{2}\right)} \quad (8)$$

Question 1(e): What is deep bar rotor induction motor? How it produces high starting torque?

SR Sir's Thought Process

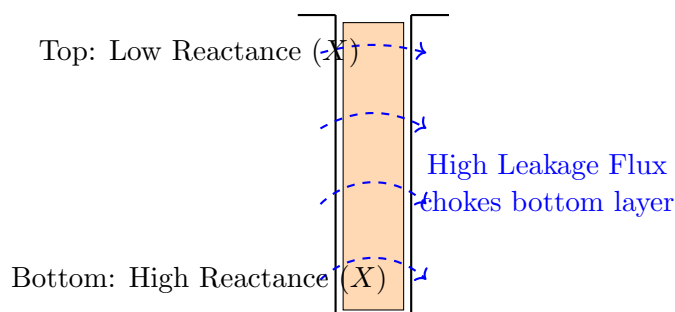
This question has appeared consecutively in 2022, 2023, and now 2024. It tests the concept of the Skin Effect in AC machinery. I will clearly differentiate between the $s = 1$ (starting) and $s \approx 0$ (running) conditions to explain the dynamic shifting of rotor resistance.

Answer:

Deep Bar Rotor Induction Motor: A deep bar rotor is a specialized squirrel cage rotor designed to overcome the classic trade-off between starting torque and running efficiency. Instead of utilizing standard shallow conductors, the rotor slots are punched deeply into the iron core, and the conductive bars (copper or aluminum) inserted into these slots possess a very high depth-to-width ratio.

Mechanism of Producing High Starting Torque: The operational magic of a deep bar rotor relies exclusively on the **Skin Effect**, which dynamically alters the effective resistance based on the slip frequency of the rotor currents.

Consider a single deep rotor bar divided conceptually into a top layer (near the air gap) and a bottom layer (deep in the iron core). The bottom layer is surrounded by a massive amount of magnetic iron, meaning it links with a much larger amount of local slot leakage flux compared to the top layer. Because inductance is proportional to flux linkage, the bottom layer inherently has a much higher leakage inductance (L) and leakage reactance ($X = 2\pi fL$) than the top layer.



- At Starting ($s = 1$):** The rotor is stationary. Therefore, the frequency of the induced rotor currents is identical to the main supply frequency (e.g., 50 Hz). At this high frequency, the reactance ($2\pi fL$) of the bottom portion of the bar becomes so overwhelmingly large that the current is literally choked off and forced to flow entirely through the narrow top layer (Skin Effect). This severe crowding reduces the effective cross-sectional area of the conductor, resulting in a very high effective rotor resistance ($R_{2,eff}$). Since starting torque is directly proportional to rotor resistance ($T_{st} \propto R_{2,eff}$), the motor successfully develops a massive starting torque.
- At Running Condition ($s \approx 0.03$):** As the motor approaches synchronous speed, the slip frequency drops drastically to around 1.5 Hz. At this extremely low frequency, the frequency-dependent reactance term (X) vanishes almost entirely. The current is no longer choked and distributes evenly across the entire depth of the deep bar, utilizing the full cross-sectional area. The effective resistance plummets to its normal DC value, ensuring minimal I^2R copper losses and excellent running efficiency.

Question 2(a): Explain how a rotating magnetic field with constant amplitude is generated in an three phase induction machine.

SR Sir's Thought Process

This requires a rigorous mathematical proof. Three balanced spatial windings, three balanced sinusoidal time currents. Apply trigonometric identities to show the backward-rotating components sum to zero, leaving a purely forward-rotating wave with amplitude $1.5F_m$.

Answer:

A Rotating Magnetic Field (RMF) of constant amplitude is generated naturally when a balanced 3-phase AC voltage is applied to a balanced 3-phase distributed spatial winding.

Mathematical Explanation: Let the stator hold three identical windings (Phases A, B, and C) whose magnetic axes are displaced from one another in space by 120° electrical. When connected to a 3-phase supply, the instantaneous currents flowing through these windings are displaced in time by 120° :

$$i_A(t) = I_m \cos(\omega t) \quad (9)$$

$$i_B(t) = I_m \cos(\omega t - 120^\circ) \quad (10)$$

$$i_C(t) = I_m \cos(\omega t + 120^\circ) \quad (11)$$

Each phase current produces a pulsating Magnetomotive Force (MMF) along its own fixed spatial axis. Let θ be the arbitrary spatial angle measured from the axis of Phase A. The spatial MMFs generated by each phase at any point θ in the air gap are:

$$F_A(\theta, t) = F_m \cos(\omega t) \cos(\theta) \quad (12)$$

$$F_B(\theta, t) = F_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ) \quad (13)$$

$$F_C(\theta, t) = F_m \cos(\omega t + 120^\circ) \cos(\theta + 120^\circ) \quad (14)$$

where F_m is the peak MMF of a single individual phase.

The total resultant MMF in the air gap (F_{res}) is the algebraic sum of these three pulsating fields. We apply the trigonometric identity $\cos X \cos Y = \frac{1}{2}[\cos(X - Y) + \cos(X + Y)]$ to expand each term:

$$F_A = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta)] \quad (15)$$

$$F_B = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta - 240^\circ)] \quad (16)$$

$$F_C = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta + 240^\circ)] \quad (17)$$

Now, sum the three expanded equations:

$$\begin{aligned} F_{res}(\theta, t) &= F_A + F_B + F_C \\ &= \frac{F_m}{2} [3 \cos(\omega t - \theta)] + \frac{F_m}{2} [\cos(\omega t + \theta) + \cos(\omega t + \theta - 240^\circ) + \cos(\omega t + \theta + 240^\circ)] \end{aligned} \quad (18)$$

Notice the second bracket containing the $(\omega t + \theta)$ terms. These represent backward-traveling waves. Because they are separated exactly by 240° (which is functionally identical to $\pm 120^\circ$), they form a perfectly balanced 3-phase vector set. The sum of any three balanced vectors is always zero. Thus, the entire backward-traveling component perfectly cancels out.

We are left with only the forward-traveling component:

$$\boxed{F_{res}(\theta, t) = \frac{3}{2}F_m \cos(\omega t - \theta)} \quad (19)$$

This final equation definitively proves that the resultant magnetic field has a **constant amplitude** of $1.5F_m$ and rotates at a constant synchronous angular velocity ω .

Question 2(b) [Printed as 2(a)]: Derive an expression for developed torque of an induction motor. Also derive the value of slip for which torque is maximum and draw the torque-speed characteristics.

SR Sir's Thought Process

Derive the torque equation from air-gap power (P_{ag}). Find maximum torque slip by differentiating with respect to slip s . Plot the characteristic curve.

Answer:

Derivation of Developed Torque (T_e): From the per-phase exact equivalent circuit of the induction motor referred to the stator side, the rotor current I'_2 at any slip s is given by:

$$I'_2 = \frac{V_1}{\sqrt{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2}} \quad (20)$$

The gross power transferred across the air gap from the stator to the rotor (P_{ag}) is dissipated entirely in the effective rotor resistance branch. For all 3 phases:

$$P_{ag} = 3(I'_2)^2 \frac{R'_2}{s} \quad (21)$$

Electromagnetic torque (T_e) is mathematically defined as the air gap power divided by the synchronous speed in mechanical radians per second ($\omega_s = \frac{2\pi N_s}{60}$):

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{3}{\omega_s} (I'_2)^2 \frac{R'_2}{s} \quad (22)$$

Substituting the expression for I'_2 from (20) yields the rigorous torque equation:

$$T_e = \frac{3}{\omega_s} \frac{V_1^2 \left(\frac{R'_2}{s}\right)}{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2} \text{ N-m} \quad (23)$$

Slip for Maximum Torque (s_{mT}): To find the slip at which this torque peaks, we apply the Maximum Power Transfer Theorem. Maximum power is transferred to the variable component $\left(\frac{R'_2}{s}\right)$ when it equals the magnitude of the remaining source impedance looking back into the circuit.

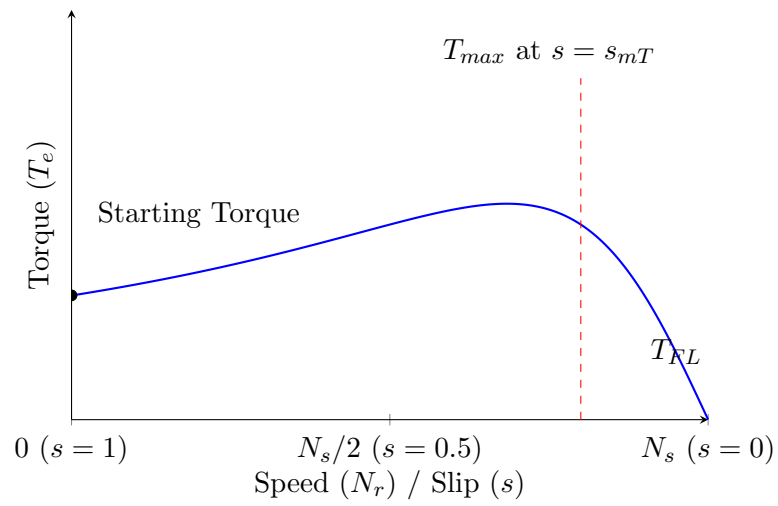
$$\frac{R'_2}{s_{mT}} = \sqrt{R_1^2 + (X_1 + X'_2)^2} \quad (24)$$

Solving for s_{mT} gives the exact slip for maximum torque:

$$s_{mT} = \frac{R'_2}{\sqrt{R_1^2 + (X_1 + X'_2)^2}} \quad (25)$$

Note: In simplified analyses where stator resistance R_1 is neglected, this collapses to $s_{mT} \approx \frac{R'_2}{X_1 + X'_2}$.

Torque-Speed Characteristics:



Question 3(a): From the basic principle develop the equivalent circuit of a poly phase induction motor.

SR Sir's Thought Process

1. Start with the transformer principle. Draw stator and rotor as separate coupled circuits. 2. Establish equations: Rotor induced voltage $E_{2s} = sE_2$. Rotor reactance $X_{2s} = sX_2$. 3. Show current $I_2 = \frac{sE_2}{\sqrt{R_2^2 + s^2 X_2^2}}$. 4. Divide by s to decouple frequency, creating a fictitious stationary secondary: $I_2 = \frac{E_2}{\sqrt{(R_2/s)^2 + X_2^2}}$. 5. Refer values to stator side and construct the final diagram.

Answer:

An induction motor operates on the exact principles of a transformer with a short-circuited, rotating secondary. Let V_1 be the per-phase supply voltage. The stator winding has resistance R_1 and leakage reactance X_1 . It draws a no-load current to establish the working flux, modeled by a core loss resistance R_c and magnetizing reactance X_m in parallel.

Step 1: The Rotating Rotor Circuit At standstill ($s = 1$), let the induced EMF per phase in the rotor be E_2 and its leakage reactance be X_2 . When the rotor spins at slip s , the relative speed of the magnetic field cutting the rotor conductors drops. Consequently:

- The induced rotor voltage scales with slip: $E_{2s} = sE_2$
- The rotor frequency scales with slip: $f_2 = sf_1$
- The rotor leakage reactance scales with slip: $X_{2s} = 2\pi(sf_1)L_2 = sX_2$

The rotor current (I_2) flowing through the actual ohmic resistance R_2 is:

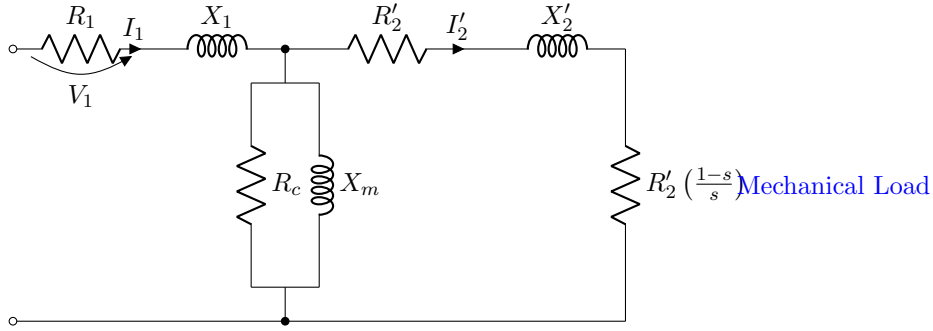
$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \quad (26)$$

Step 2: The Fictitious Stationary Rotor Equation (26) represents a rotating circuit with variable frequency. We can transform this into a mathematically equivalent stationary circuit driven by the full supply frequency. Divide the numerator and the denominator of the current equation by s :

$$I_2 = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \quad (27)$$

This pivotal mathematical trick converts the rotating secondary into a purely stationary secondary where the voltage is fixed at E_2 , the reactance is fixed at X_2 , and the resistance becomes a variable parameter $\frac{R_2}{s}$.

Step 3: Referring to the Stator To link the stator and rotor circuits, we refer the stationary rotor parameters back to the primary stator side using the effective turns ratio ($k = N_1/N_2$). $R'_2 = k^2 R_2$, $X'_2 = k^2 X_2$, and $I'_2 = I_2/k$. Finally, the variable resistance is split to isolate the physical copper loss from the mechanical load: $\frac{R'_2}{s} = R'_2 + R'_2 \left(\frac{1-s}{s}\right)$. The latter term represents the mechanical power output. This yields the exact equivalent circuit:



Question 3(b): How can you determine the equivalent circuit parameters in the laboratory?

SR Sir's Thought Process

Three standard laboratory tests are required to extract all parameters of the equivalent circuit. I need to list them and show the formulas used to calculate $R_1, R_c, X_m, R_2', X_2'$.

Answer:

The parameters of the exact equivalent circuit ($R_1, X_1, R_c, X_m, R_2', X_2'$) can be determined systematically in the laboratory using three standard tests:

1. DC Resistance Test (Determines R_1) A low voltage DC supply is connected across any two stator terminals. The resulting DC current (I_{dc}) and voltage drop (V_{dc}) are measured. For a star-connected stator, the resistance measured is across two phases. Hence, the DC resistance per phase is $R_{dc} = \frac{V_{dc}}{2I_{dc}}$. To account for the skin effect present under AC operation, this value is scaled up by a factor (usually 1.2 to 1.5): $R_1 \approx 1.2R_{dc}$.

2. No-Load Test (Determines R_c and X_m) The motor is allowed to run freely without any mechanical load connected to the shaft. Rated 3-phase AC voltage (V_0) at rated frequency is applied. Because the motor runs very close to synchronous speed, the slip $s \approx 0$. The mechanical load equivalent resistance $R_2'(\frac{1-s}{s})$ approaches infinity, effectively acting as an open circuit for the rotor branch. *Measurements:* Line voltage V_0 , Total No-Load Power P_0 , Line current I_0 . *Calculations (per phase):* Phase voltage $V_{0ph} = V_0/\sqrt{3}$. Phase power $P_{0ph} = P_0/3$. No-load power factor: $\cos \phi_0 = \frac{P_{0ph}}{V_{0ph}I_0}$. Core loss current component: $I_c = I_0 \cos \phi_0$. Magnetizing current component: $I_m = I_0 \sin \phi_0$. Shunt parameters:

$$R_c = \frac{V_{0ph}}{I_c} \quad \text{and} \quad X_m = \frac{V_{0ph}}{I_m} \quad (28)$$

3. Blocked Rotor Test (Determines R_2' and X_{eq}) The rotor shaft is physically clamped so it cannot rotate, forcing the slip to be $s = 1$. A vastly reduced 3-phase AC voltage (V_{br}) is applied gradually until exactly rated full-load current (I_{br}) flows in the stator. Because the voltage is very low, the core losses and magnetizing current are negligible, meaning the shunt branch (R_c, X_m) can be ignored. *Measurements:* Line voltage V_{br} , Total Blocked Rotor Power P_{br} , Line current I_{br} . *Calculations (per phase):* Phase voltage $V_{br,ph} = V_{br}/\sqrt{3}$. Phase power $P_{br,ph} = P_{br}/3$. Equivalent impedance: $Z_{br} = \frac{V_{br,ph}}{I_{br}}$. Equivalent resistance: $R_{br} = \frac{P_{br,ph}}{I_{br}^2}$. Equivalent reactance: $X_{br} = \sqrt{Z_{br}^2 - R_{br}^2}$. From these, we isolate the rotor parameters: Rotor resistance referred to stator: $R_2' = R_{br} - R_1$. Total leakage reactance: $X_{br} = X_1 + X_2'$. Usually, it is assumed they are equal, so $X_1 = X_2' = X_{br}/2$.

Question 4(a): i) If the stator impedance of an induction motor is neglected show that:

$$T_e/T_{em} = 2/(s_{mT}/s + s/s_{mT})$$

ii) From the equivalent circuit of a poly phase induction motor, obtain the following relations:

$$1) I_{2st}/I_2 = \sqrt{[(s^2 + s_{mT}^2)/(s^2(1 + s_{mT}^2))]}$$

$$2) I_{2mT}/I_2 = \sqrt{[0.5(1 + (s_{mT}/s)^2)]}$$

SR Sir's Thought Process

This exact derivation suite is a staple of JU exams. I will execute the precise algebraic manipulations required to strip away R_2 and X_2 leaving only ratios of slip.

Answer:

i) Proof of Torque Ratio: If stator impedance is neglected ($R_1 \approx 0, X_1 \approx 0$), the torque at any slip s is:

$$T_e = \frac{kV_1^2 \frac{R_2}{s}}{\left(\frac{R_2}{s}\right)^2 + X_2^2} = \frac{kV_1^2 s R_2}{R_2^2 + s^2 X_2^2} \quad (29)$$

The slip at maximum torque (s_{mT}) occurs when $R_2/s = X_2$, therefore $s_{mT} = \frac{R_2}{X_2} \implies X_2 = \frac{R_2}{s_{mT}}$. Substitute $s = s_{mT}$ into the torque equation to define the maximum torque T_{em} :

$$T_{em} = \frac{kV_1^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 X_2^2} = \frac{kV_1^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 \left(\frac{R_2}{s_{mT}}\right)^2} = \frac{kV_1^2 s_{mT}}{2R_2} \quad (30)$$

Now, evaluate the ratio $\frac{T_e}{T_{em}}$:

$$\frac{T_e}{T_{em}} = \left(\frac{kV_1^2 s R_2}{R_2^2 + s^2 X_2^2} \right) \div \left(\frac{kV_1^2 s_{mT}}{2R_2} \right) = \frac{2sR_2^2}{s_{mT}(R_2^2 + s^2 X_2^2)} \quad (31)$$

Divide the numerator and denominator by R_2^2 :

$$\frac{T_e}{T_{em}} = \frac{2s}{s_{mT} \left(1 + s^2 \left(\frac{X_2}{R_2} \right)^2 \right)} \quad (32)$$

Since $\frac{X_2}{R_2} = \frac{1}{s_{mT}}$, we substitute this in:

$$\frac{T_e}{T_{em}} = \frac{2s}{s_{mT} \left(1 + \frac{s^2}{s_{mT}^2} \right)} = \frac{2s}{s_{mT} + \frac{s^2}{s_{mT}}} \quad (33)$$

Divide the numerator and denominator by s :

$$\boxed{\frac{T_e}{T_{em}} = \frac{2}{\frac{s_{mT}}{s} + \frac{s}{s_{mT}}}} \quad (\text{Proved}) \quad (34)$$

ii) Proof of Current Relations: The generalized expression for rotor current at slip s is:

$$I_2 = \frac{V_1}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} = \frac{sV_1}{\sqrt{R_2^2 + s^2 X_2^2}} \quad (35)$$

Substituting $R_2 = s_{mT}X_2$:

$$I_2 = \frac{sV_1}{\sqrt{s_{mT}^2 X_2^2 + s^2 X_2^2}} = \frac{sV_1}{X_2 \sqrt{s_{mT}^2 + s^2}} \quad (36)$$

Relation 1: Ratio of Starting Current to Running Current At starting, $s = 1$. Using (36), the starting current I_{2st} is:

$$I_{2st} = \frac{(1)V_1}{X_2 \sqrt{s_{mT}^2 + 1^2}} = \frac{V_1}{X_2 \sqrt{s_{mT}^2 + 1}} \quad (37)$$

Divide I_{2st} by I_2 :

$$\frac{I_{2st}}{I_2} = \frac{\frac{V_1}{X_2 \sqrt{s_{mT}^2 + 1}}}{\frac{sV_1}{X_2 \sqrt{s_{mT}^2 + s^2}}} = \frac{\sqrt{s^2 + s_{mT}^2}}{s \sqrt{1 + s_{mT}^2}} = \sqrt{\frac{s^2 + s_{mT}^2}{s^2(1 + s_{mT}^2)}} \quad (38)$$

$$\boxed{\frac{I_{2st}}{I_2} = \sqrt{\frac{s^2 + s_{mT}^2}{s^2(1 + s_{mT}^2)}}} \quad (\text{Proved}) \quad (39)$$

Relation 2: Ratio of Max Torque Current to Running Current At maximum torque, $s = s_{mT}$. Using (36), the current I_{2mT} is:

$$I_{2mT} = \frac{s_{mT}V_1}{X_2 \sqrt{s_{mT}^2 + s_{mT}^2}} = \frac{s_{mT}V_1}{X_2 \sqrt{2s_{mT}^2}} = \frac{V_1}{X_2 \sqrt{2}} \quad (40)$$

Divide I_{2mT} by I_2 :

$$\frac{I_{2mT}}{I_2} = \frac{\frac{V_1}{X_2 \sqrt{2}}}{\frac{sV_1}{X_2 \sqrt{s^2 + s_{mT}^2}}} = \frac{\sqrt{s^2 + s_{mT}^2}}{s \sqrt{2}} = \sqrt{\frac{s^2 + s_{mT}^2}{2s^2}} \quad (41)$$

$$\frac{I_{2mT}}{I_2} = \sqrt{\frac{1}{2} \left(\frac{s^2}{s^2} + \frac{s_{mT}^2}{s^2} \right)} \quad (42)$$

$$\boxed{\frac{I_{2mT}}{I_2} = \sqrt{0.5 \left(1 + \left(\frac{s_{mT}}{s} \right)^2 \right)}} \quad (\text{Proved}) \quad (43)$$

Question 4(b): Describe a double cage induction motor. Draw and explain the equivalent circuit of double cage rotor induction motor.

SR Sir's Thought Process

Double cage rotors solve the starting vs running trade-off physically rather than mathematically. I need to describe the inner cage (low R , high X) and outer cage (high R , low X), explain how frequency controls current division, and draw the parallel equivalent circuit.

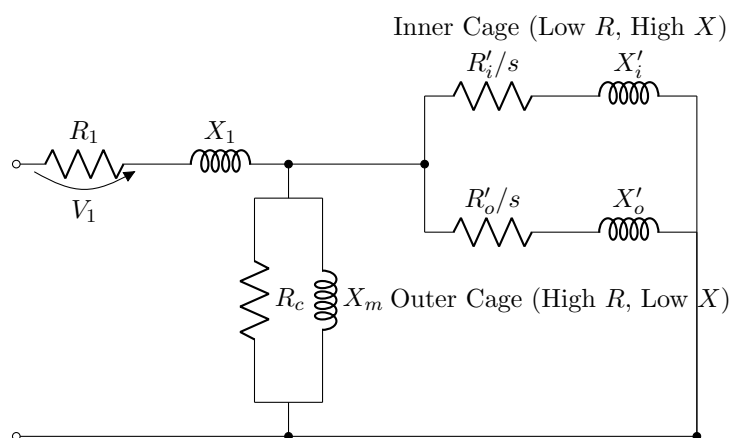
Answer:

Description of Double Cage Induction Motor: A double cage induction motor features a specialized rotor core containing two independent squirrel cages embedded in the same physical iron structure, separated radially.

- **Outer Cage:** Positioned near the air gap. It is typically made of a high-resistivity material like brass or bronze. Because it is close to the surface, it links with very little slot leakage flux. Therefore, it possesses **High Resistance (R_o)** and **Low Reactance (X_o)**.
- **Inner Cage:** Positioned deeper inside the iron core. It is made of low-resistivity pure copper. Because it is deeply embedded in the magnetic iron, it links with a massive amount of leakage flux. Therefore, it possesses **Low Resistance (R_i)** and **High Reactance (X_i)**.

During starting ($s = 1$, high frequency), the inner cage's high reactance aggressively chokes the current. The current is forced into the high-resistance outer cage, generating excellent starting torque. During running ($s \approx 0$, very low frequency), reactances become negligible. Current follows the path of least DC resistance, shifting predominantly into the low-resistance inner cage, ensuring high efficiency.

Equivalent Circuit of Double Cage Rotor: Because the two cages are effectively physically shorted together by the end rings, they act as two distinct parallel electrical circuits connected across the same induced rotor voltage. This parallel nature is directly reflected in the equivalent circuit by having two distinct rotor branches in parallel.



Explanation of Circuit Components:

- R'_o and X'_o : Effective resistance and leakage reactance of the outer cage, referred to the stator.

- R'_i and X'_i : Effective resistance and leakage reactance of the inner cage, referred to the stator.
- The total rotor current divides between the two branches depending entirely on the slip s . At start, the high value of X'_i dominates, pushing current to the R'_o branch. At run, the $1/s$ multiplier makes the resistances massive, but R'_i/s is much smaller than R'_o/s , drawing the current there.

Question 5(a): The rotor resistance and standstill reactance of a 3-phase induction motor are 0.012 ohm and 0.08 ohm per phase respectively. The full load slip is 4% at normal voltage. Calculate the percentage reduction in stator voltage to develop the full-load torque at 75% of full-load speed.

SR Sir's Thought Process

This requires setting up a torque balance equation. Let T_1 be torque at normal V_1 . Let T_2 be torque at reduced V_2 . We want $T_2 = T_{fl}$. Calculate full-load speed ($0.96N_s$), then find the new target speed ($0.75 \times 0.96 = 0.72N_s$). From this, calculate the new slip ($s_2 = 0.28$). Then use $T \propto \frac{V^2 s R_2}{R_2^2 + s^2 X_2^2}$ to solve for $(V_2/V_1)^2$.

Answer:

Given Data: Rotor resistance $R_2 = 0.012 \Omega$ Rotor standstill reactance $X_2 = 0.08 \Omega$ Full-load slip at normal voltage $s_1 = 4\% = 0.04$

Step 1: Establishing the Initial State (Full Load Torque) Assuming stator impedance is negligible for this proportional calculation, the electromagnetic torque is given by:

$$T_e = K \frac{V_1^2 \cdot s_1 \cdot R_2}{R_2^2 + s_1^2 X_2^2} \quad (44)$$

Let T_{fl} be the full load torque developed at normal voltage V_1 :

$$T_{fl} = KV_1^2 \frac{0.04 \times 0.012}{(0.012)^2 + (0.04 \times 0.08)^2} \quad (45)$$

$$= KV_1^2 \frac{0.00048}{0.000144 + 0.0001024} \quad (46)$$

$$= KV_1^2 \frac{0.00048}{0.00015424} = 3.11203 \cdot KV_1^2 \quad (47)$$

Step 2: Determining the New Operating Slip (s_2) Let N_s be the synchronous speed. The initial full-load speed is:

$$N_{fl} = (1 - s_1)N_s = (1 - 0.04)N_s = 0.96N_s \quad (48)$$

The motor is required to operate at 75% of this full-load speed:

$$N_2 = 0.75 \times N_{fl} = 0.75 \times (0.96N_s) = 0.72N_s \quad (49)$$

The required slip (s_2) corresponding to this new speed is:

$$s_2 = \frac{N_s - N_2}{N_s} = \frac{N_s - 0.72N_s}{N_s} = 0.28 \text{ (or 28\%)} \quad (50)$$

Step 3: Calculating Required Voltage (V_2) At this new slip s_2 , the torque developed must equal the original full-load torque T_{fl} .

$$T_2 = K \frac{V_2^2 \cdot s_2 \cdot R_2}{R_2^2 + s_2^2 X_2^2} \quad (51)$$

$$= KV_2^2 \frac{0.28 \times 0.012}{(0.012)^2 + (0.28 \times 0.08)^2} \quad (52)$$

$$= KV_2^2 \frac{0.00336}{0.000144 + 0.00050176} \quad (53)$$

$$= KV_2^2 \frac{0.00336}{0.00064576} = 5.20317 \cdot KV_2^2 \quad (54)$$

Equating (47) and (54) since T_2 must equal T_{fl} :

$$3.11203 \cdot KV_1^2 = 5.20317 \cdot KV_2^2 \quad (55)$$

$$\left(\frac{V_2}{V_1}\right)^2 = \frac{3.11203}{5.20317} = 0.5981 \quad (56)$$

$$\frac{V_2}{V_1} = \sqrt{0.5981} = 0.7733 \text{ (or 77.33\%)} \quad (57)$$

Step 4: Percentage Reduction The new voltage V_2 is 77.33% of the original normal voltage V_1 . The percentage reduction in stator voltage is:

$$\text{Reduction} = 100\% - 77.33\% = \mathbf{22.67\%} \quad (58)$$

Question 5(b): The standstill impedances of the two cages of a double-cage induction motor are $(3.2 + j 1.2)$ ohm and $(0.5 + j 6.5)$ ohm respectively. If the full-load slip is 5%, then find the ratio of starting torque to full-load torque. Neglect magnetizing current and stator impedance.

SR Sir's Thought Process

Calculate power transferred to the rotor at $s = 1$ (starting) and $s = 0.05$ (full load) for both parallel cages. Torque $T \propto P_{ag} \propto 3I_{branch}^2(R/s)$. Since cages are in parallel across voltage V : $P_{branch} \propto V^2 \frac{R/s}{(R/s)^2 + X^2}$. Sum the power components for both cages at $s = 1$ to get T_{st} . Sum the power components for both cages at $s = 0.05$ to get T_{fl} . Divide the two results.

Answer:

Given Data: Outer cage impedance (standstill): $Z_o = R_o + jX_o = 3.2 + j1.2 \Omega$ Inner cage impedance (standstill): $Z_i = R_i + jX_i = 0.5 + j6.5 \Omega$ Full-load slip: $s_{fl} = 5\% = 0.05$

Instructor's Note: The torque developed by an induction motor is proportional to the air-gap power, which is the sum of the power dissipated in the effective rotor resistances (R/s) of both parallel cages. Because stator impedance is neglected, the full supply voltage V is applied directly across the parallel rotor branches. The torque developed by any single cage is:

$$T_{cage} \propto \frac{V^2 \left(\frac{R}{s}\right)}{\left(\frac{R}{s}\right)^2 + X^2} \quad (59)$$

Total torque is the sum of the torques developed by the inner and outer cages.

Step 1: Calculate Starting Torque (T_{st}) Proportionality At starting, $s = 1$. Torque of outer cage ($T_{o,st}$):

$$T_{o,st} \propto \frac{V^2(3.2)}{(3.2)^2 + (1.2)^2} = V^2 \frac{3.2}{10.24 + 1.44} = V^2 \frac{3.2}{11.68} = 0.27397V^2 \quad (60)$$

Torque of inner cage ($T_{i,st}$):

$$T_{i,st} \propto \frac{V^2(0.5)}{(0.5)^2 + (6.5)^2} = V^2 \frac{0.5}{0.25 + 42.25} = V^2 \frac{0.5}{42.50} = 0.01176V^2 \quad (61)$$

Total starting torque:

$$T_{st} \propto 0.27397V^2 + 0.01176V^2 = \mathbf{0.28573V^2} \quad (62)$$

Step 2: Calculate Full-Load Torque (T_{fl}) Proportionality At full load, $s = 0.05$. The effective resistances become: $R_{o,eff} = \frac{R_o}{s} = \frac{3.2}{0.05} = 64 \Omega$ $R_{i,eff} = \frac{R_i}{s} = \frac{0.5}{0.05} = 10 \Omega$

Torque of outer cage ($T_{o,fl}$):

$$T_{o,fl} \propto \frac{V^2(64)}{(64)^2 + (1.2)^2} = V^2 \frac{64}{4096 + 1.44} = V^2 \frac{64}{4097.44} = 0.01562V^2 \quad (63)$$

Torque of inner cage ($T_{i,fl}$):

$$T_{i,fl} \propto \frac{V^2(10)}{(10)^2 + (6.5)^2} = V^2 \frac{10}{100 + 42.25} = V^2 \frac{10}{142.25} = 0.07030V^2 \quad (64)$$

Total full-load torque:

$$T_{fl} \propto 0.01562V^2 + 0.07030V^2 = \mathbf{0.08592V^2} \quad (65)$$

Step 3: Ratio of Starting Torque to Full-Load Torque

$$\frac{T_{st}}{T_{fl}} = \frac{0.28573V^2}{0.08592V^2} = \mathbf{3.325} \quad (66)$$

The ratio of starting torque to full-load torque is **3.325 : 1**.

— **END OF 2024 PART I SOLUTIONS** —
May you master these concepts with absolute clarity.